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Demonstration of a wireless, self-powered, electroacoustic liner system^{a)}

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This paper demonstrates the system operation of a self-powered active liner for the suppression of aircraft engine noise. The fundamental element of the active liner system is an electromechanical Helmholtz resonator (EMHR), which consists of a Helmholtz resonator with one of its rigid walls replaced with a circular piezoceramic composite plate. For this system demonstration, two EMHR elements are used, one for acoustic impedance tuning and one for energy harvesting. The EMHR used for acoustic impedance tuning is shunted with a variable resistive load, while the EMHR used for energy harvesting is shunted to a flyback power converter and storage element. The desired acoustic impedance conditions are determined externally, and wirelessly transmitted to the liner system. The power for the receiver and the impedance tuning circuitry in the liner are supplied by the harvested energy. Tuning of the active liner is demonstrated at three different sound pressure levels (148, 151, and 153 dB) in order to show the robustness of the energy harvesting and storage system. An acoustic tuning range of approximately 200 Hz is demonstrated for each of the three available power levels. © 2009 Acoustical Society of America. [DOI: 10.1121/1.3050287]

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I. INTRODUCTION

Noise suppression provided by acoustic liners in aircraft engine ducts is necessary for modern commercial aircraft to meet Federal Aviation Administration flyover noise certifications.^{1,2} There are two categories of acoustic liners used to reduce engine noise: passive and active. Passive acoustic liners usually employ a single layer system consisting of a honeycomb septum sandwiched between a solid backplate and a perforated facesheet that provides complex impedance boundary conditions for the noise propagation along the engine duct.¹ The advantage of passive liners is their simplicity and structural integrity, which leads to relatively low-cost fabrication and implementation. However, such systems are only capable of operating over a relatively narrow frequency range. For instance, the useful bandwidth of a single degree-of-freedom liner is approximately one octave.¹ Although it is possible to extend the frequency range of noise reduction of passive liners by adopting a multilayer structure, the weight and size requirements are often prohibi-

tive. Moreover, the performance characteristics of a passive acoustic liner are essentially fixed, except for flow and acoustic level dependence.¹

Active or adaptive liners, on the other hand, offer the promise of *in situ* adjustment of liner performance over a broad noise frequency range. Generally, active or adaptive liners are systems that employ sensors, mechanical actuators, and controllers to modify the geometry of the liner or use bias flow through the perforated sheet.^{3–6} However, the sensors and actuators add size and complexity to the liner system, making design and implementation difficult.

In order to achieve *in situ* adjustment of the liner without the use of mechanical actuators, a novel method using electromechanical Helmholtz resonators (EMHRs) has been reported by Horowitz *et al.*⁷ and Liu *et al.*⁸ An EMHR, shown in Fig. 1, is comprised of an orifice or neck, backing cavity, and a compliant, composite piezoelectric backplate that is attached to an electrical shunt network. By modifying the shunt network, the acoustic impedance of the EMHR and its resonant frequencies can be tuned *in situ*.^{7,8}

Similar to all active tuning methods, control signals and power must be provided for the tuning circuitry of the EMHR. The control signals can be transmitted wirelessly to EMHR tuning networks, which eliminates electrical wiring but does not alleviate the power requirements. An alternative to providing power from an external source is to configure

^{a)}Preliminary portions of this work were presented in “A self-powered wireless active acoustic liner,” 12th AIAA/CEAS Aeroacoustics Conference, Cambridge, MA, May 2006, AIAA Paper No. 2006-2400 and “Technology development for electromechanical acoustic liners,” Active 04, Williamsburg, VA, May 2004.

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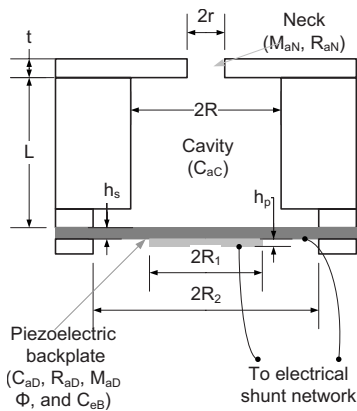


FIG. 1. Cross section of an EMHR.

the system to be self-powered. A self-powered system is one in which all of the required operating energy can be harvested from the environment.^{9–11} Recently, it was shown by Liu *et al.*⁸ that energy harvesting in an acoustic environment can be achieved with the same EMHR geometry used for acoustic impedance tuning.¹² Thus, a wireless self-powered active liner system can be realized by employing an array of EMHR elements with a common geometry. Figure 2 shows one potential configuration in which EMHR elements can be used to either adjust the local acoustic impedance boundary condition or to harvest the energy needed to operate the liner.

This paper presents the integration and experimental verification of an acoustically self-powered active liner system, expanding on the previous preliminary work reported by Kadirvel *et al.*¹³ In the previous conference paper, results on the operation of each *isolated* component were presented. In this paper, the frequency tuning of the acoustically self-powered active liner is characterized in detail as a whole, verifying interoperability of the entire system. Three different acoustic energy harvesting scenarios are employed to demonstrate system robustness and sustainability with different amounts of available acoustic power.

The paper is organized as follows. Section II examines the operation of the active liner system. The complete integrated system is presented first, followed by a detailed discussion of each system block. Section III describes the specific components used for implementation of the prototype

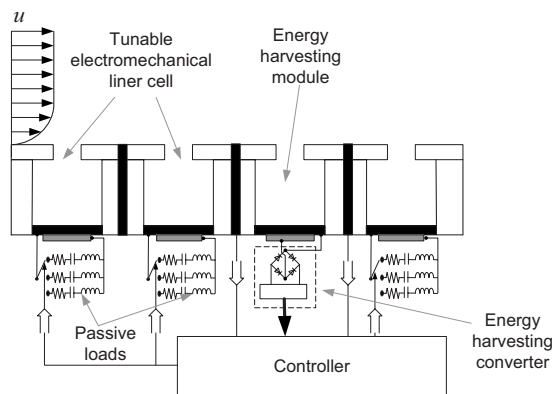


FIG. 2. Active liner composed of an array of two types of EMHR elements, one to tune the acoustic impedance of the liner and the other to harvest acoustic energy.

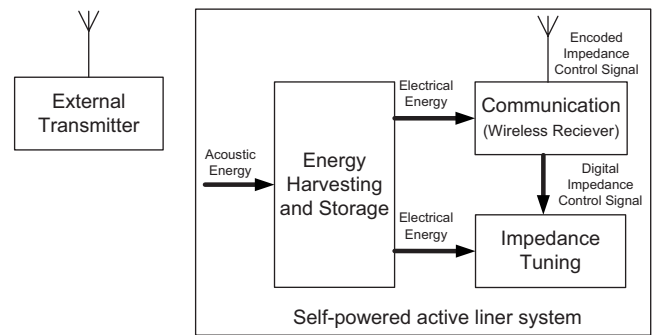


FIG. 3. Active liner system with external transmitter and various blocks for impedance tuning, energy harvesting, and communication.

and the methods employed for experimental verification. Experimental results are presented in Sec. IV, and conclusions and future work are provided in Sec. V.

II. EMHR-ARRAY ACTIVE LINER SYSTEM

The active liner system in this paper is operated in an open-loop configuration to reduce complexity and minimize the energy requirements of the liner. Closed-loop systems require output detection and feedback circuitry, both of which increase the system complexity and size, and possibly the power requirements.

The self-powered active liner system presented in this work can be divided into three functional blocks: impedance tuning, energy harvesting, and communication. A system diagram, shown in Fig. 3, illustrates how the blocks interact to tune the impedance boundary conditions inside the duct. The impedance tuning block is comprised of components that physically adjust the local impedance boundary condition of the duct. In this open-loop implementation, the control signal to set the desired boundary conditions is wirelessly transmitted from an external transmitter in the form of the impedance control signal (ICS). The ICS is received by the communication block where it is decoded, and is used by the impedance tuning block to set the desired acoustic impedance. Incident acoustic energy is converted into electrical energy within the energy harvesting block, where it is used to power the other liner blocks or is stored for future use.

A. Impedance tuning block

The EMHR element is the fundamental structure within the impedance tuning block for adjusting the acoustic boundary conditions for the noise propagation along the duct. The EMHR is a multiple energy domain device, which couples the acoustic and electrical domains via the piezoelectric composite backplate. A lumped element model (LEM) of the EMHR was developed by Liu *et al.*⁸ The LEM, shown in Fig. 4, models the parameters of the EMHR as discrete cir-

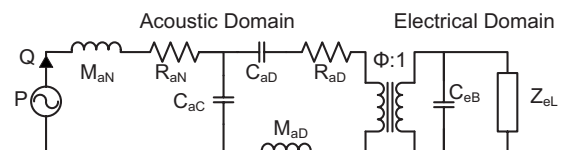


FIG. 4. LEM of the EMHR.

TABLE I. LEM parameters for a single EMHR element.

LEM parameter	Definition
P	Incident acoustic pressure
Q	Incident volumetric flow
M_{aN}	Neck acoustic mass
R_{aN}	Neck acoustic damping loss
C_{aC}	Cavity acoustic compliance
C_{aD}	Diaphragm acoustic compliance
M_{aD}	Diaphragm acoustic mass
R_{aD}	Diaphragm acoustic damping loss
ϕ	Piezoelectric transduction ratio
C_{eB}	Blocked electrical capacitance

cuit elements connected to a passive electrical load, Z_{eL} . The definition of each element in the LEM is provided in Table I. Unlike a standard Helmholtz resonator, whose acoustic impedance is only a function of the device geometry, the acoustic impedance of the EMHR is also a function of its electrical load. The acoustic input impedance of an EMHR is given by⁸

$$Z_{aIN} = R_{aN} + sM_{aN} + \frac{1}{sC_{aC} \left(\frac{1}{sC_{aD}} + R_{aD} + sM_{aD} + \frac{\phi^2 Z_{eL}}{1 + sC_{eB}Z_{eL}} \right)} + \frac{1}{sC_{aC} + \frac{1}{sC_{aD}} + R_{aD} + sM_{aD} + \frac{\phi^2 Z_{eL}}{1 + sC_{eB}Z_{eL}}} \Bigg|_{s=j\omega} \quad (1)$$

where $s=j\omega$ and ω is the angular frequency. The effects of using resistive, capacitive, or inductive electrical loads to tune the acoustic impedance were demonstrated by Liu *et al.*⁸ For illustration purposes, only purely resistive electrical loads are considered in this paper. The EMHR with a resistive load is analogous to a two degree-of-freedom system, and has two resonant frequencies [i.e., f_1 and f_2 at $\text{Im}(Z_{aIN})=0$]. Both resonant frequencies of the EMHR can be tuned simply by varying the resistance value. The tuning range of the resonant frequencies extend between the short-circuit ($Z_{eL}=0$) and open-circuit cases ($Z_{eL}=\infty$).⁸

The other element of the impedance tuning block is the switching array, which is used to vary the shunt electrical impedance across the EMHR. The switching array is comprised of N switch-load pairs connected in parallel across the EMHR. The case of purely resistive loads is shown in Fig. 5. After the ICS is received and decoded in the communication block, an N -bit digital signal is sent to the switching array. Each of the N bits has a logic value of either 0 or 1, and is used to control a specific switch. Depending on the logic value, the switch is either opened (i.e., 0) or closed (i.e., 1) to

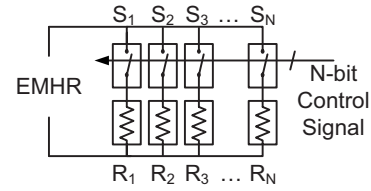


FIG. 5. Switching array in the impedance tuning block using purely resistive loads.

set the shunt impedance of the EMHR. While the ability to have multiple switches closed at the same time increases the number of possible impedance values, this work will only consider the case of any one of the N single closed switches for simplicity.

B. Energy harvesting block

The energy harvesting block is designed to provide electrical power to the active liner system. It is comprised of an EMHR element, a power converter, and a storage device. The EMHR can gather energy from the high intensity acoustic field present in the engine nacelle, which can reach levels up to 160 dB (ref. 20 μPa).¹⁴ As stated previously, the EMHR element used in this block has the same geometry as in the impedance tuning block but functions as a transducer to convert acoustic energy into electrical energy. Electrical energy harvested by the EMHR is conditioned by the power converter and is either delivered to the liner system or stored for later use. Figure 6 shows the general flow of power in the energy harvesting block. The behavior of the storage element is determined by the amount of conditioned power available from the EMHR and power converter. When the amount of conditioned power is larger than what is required by the liner electronics, excess power flows into the storage element. When the two are equal, the net power flow into the storage element is zero. For the case where the required power is larger than the available conditioned power, power flows from the storage element for sustained operation.

The role of the power converter in the energy harvesting block is to maximize the amount of electrical power delivered from the EMHR energy harvester to the load. The circuitry used in the power converter, shown in Fig. 7, is comprised of a full-bridge rectifier-capacitor circuit and a flyback converter. Operation of this circuit for acoustic energy harvesting was examined in detail by Liu *et al.*,¹² and is only briefly reviewed here. The active liner electronics requires a stable dc voltage, and the rectifier is used to condition the time varying ac signal produced by the EMHR. It was shown that for a rectifier-capacitor circuit driven by a sinusoidal source, an optimal resistive load for the rectifier exists for

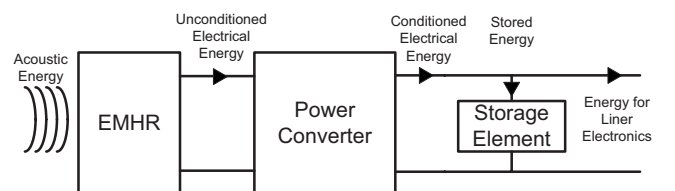


FIG. 6. Power flow diagram of the energy harvesting block.

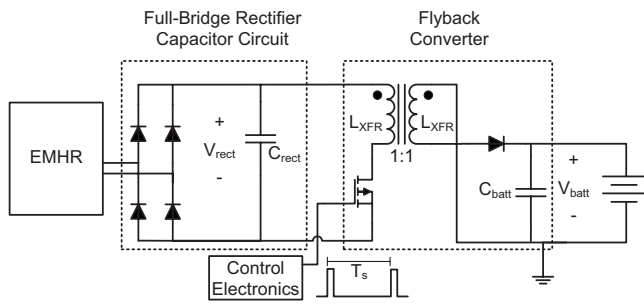


FIG. 7. Detailed diagram of the circuitry within the energy harvesting block.

which the power delivered from the EMHR to the load will be maximized.¹² The flyback converter is used to emulate this optimal loading condition for the rectifier. The input impedance of the flyback converter, operated in discontinuous conduction mode, is purely resistive and is given by¹⁵

$$R_{in} = \frac{2f_s L_{XFR}}{d^2}, \quad (2)$$

where L_{XFR} is the inductance of the flyback transformer, f_s is the switching frequency of the converter, and d is the duty cycle. Adjustment of the input impedance can be controlled electronically by varying d . Since the input impedance of the flyback is independent of its load, it is placed between the EMHR energy harvester and the liner electronics to decouple the two and provide the optimal resistance.

A rechargeable battery is chosen as the storage element in the energy harvesting block because its voltage remains approximately constant as it is charged and discharged. The constant voltage of the battery can be used to provide a stable voltage source for the liner electronics and eliminates the need for voltage regulation circuitry.

C. Wireless communication block and system integration

The desired acoustic impedance is determined externally to the active liner, as a function of engine operating conditions (e.g., take-off, cut back, and approach), and the appropriate ICS is wirelessly transmitted to the system. The communication block, comprised of the self-powered receiver and associated electronics, receives the ICS. A detailed block diagram, presented in Fig. 8, illustrates how the communication block interacts with the other system blocks. The wireless receiver is integrated directly into the liner system and receives its power from the energy harvesting block. The wireless transmitter is external to the active liner system and receives power and input signals externally.

Transmission of the ICS is accomplished by a Holtek HT-12E encoder and a Rayming TX-99 transmission module, shown in Fig. 9(a). The HT-12E encodes 12 bits of digital data for transmission: 4 bits for the ICS and 8 address bits. Each of the four ICS bits corresponds to a specific switch in the impedance tuning switching array and determines whether it will be opened or closed. The address bits allow the transmitted signal to target specific receiver nodes.

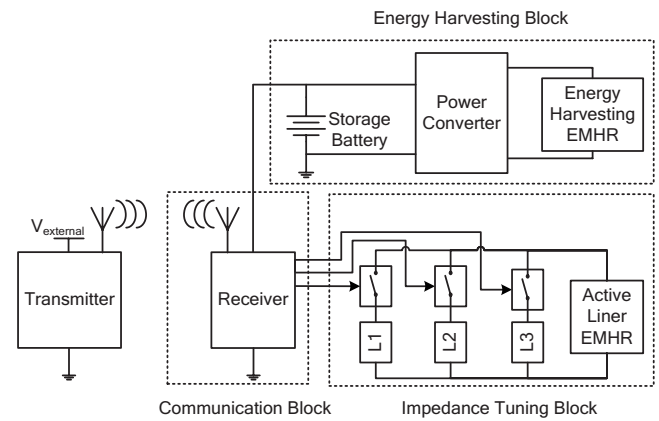


FIG. 8. Detailed block diagram of the integrated active liner system with three different resistive loads L1, L2, and L3.

Once encoded, the signal is transmitted by the TX-99 using an amplitude modulation scheme at 300 MHz with a loop antenna.

The data transmitted by the TX-99 are received at the active liner by its complementary receiver, the Rayming RX-99. The Holtek HT-12D decoder separates the 12 data bits to recover the ICS and address, as shown in Fig. 9(b). If the transmitted address bits match the preset address of the receiver, the ICS is latched in and sent to the impedance tuning block to select the appropriate load.

III. EXPERIMENTAL SETUP

Previous studies have considered the operation of the impedance tuning and energy harvesting subsystems individually.^{8,12} For the self-powered active liner system presented in this work, operation of the entire integrated system is demonstrated. In practice, a complete active liner system would be comprised of an array of EMHR elements. For proof of concept, the active liner system in this work consists of only two EMHR elements—one for noise suppression and

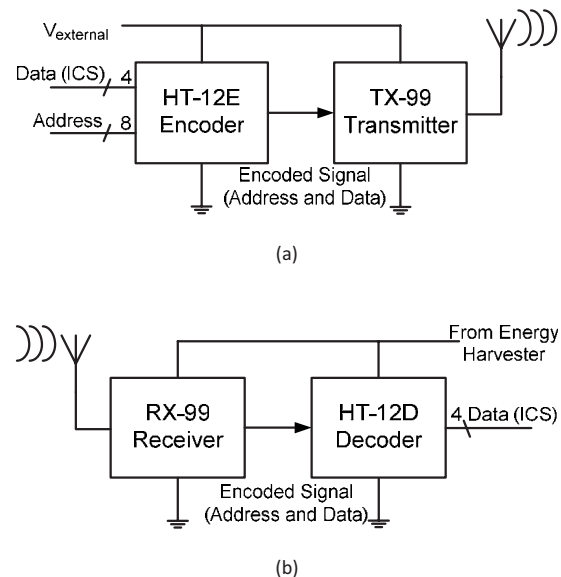


FIG. 9. Schematic of (a) the external transmitter and (b) the receiver circuitry used in the communication block.

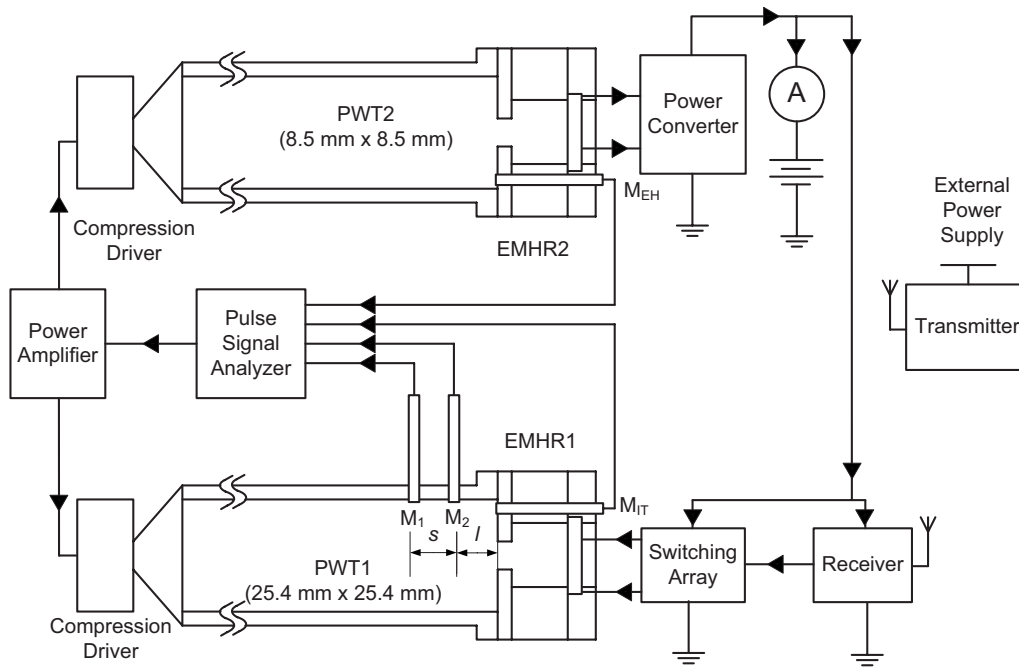


FIG. 10. Experimental setup used for self-powered active liner system testing.

one for energy harvesting—and a single receiver module. To characterize the acoustic impedance of the EMHR, the two microphone method (TMM) is used.^{16,17} The TMM is a standard technique for the determination of the normal specific acoustic impedance of acoustic samples.¹⁶ The TMM employs a plane wave tube (PWT) and uses two flush wall-mounted microphones to simultaneously measure acoustic pressure at two known positions in the tube. The acoustic sample is mounted at one end of the PWT and the sound source at the other end. The frequency response function (FRF) is computed between the two microphones using the measured pressure signals. The complex acoustic reflection coefficient is then calculated by

$$\mathfrak{R}(\omega) = \frac{\hat{H}(\omega) - e^{-jks}}{e^{jks} - \hat{H}(\omega)} e^{j2k(l+s)}, \quad (3)$$

where $\hat{H}(\omega)$ is the estimated FRF, $k = \omega/c_0$ is the wavenumber, c_0 is the isentropic speed of sound, s is the spacing between the microphone locations, and l is the distance from the sample to the nearest microphone location. The normalized specific acoustic impedance is then calculated by

$$\zeta(\omega) = \theta(\omega) + j\chi(\omega) = \frac{1 + \mathfrak{R}(\omega)}{1 - \mathfrak{R}(\omega)}, \quad (4)$$

where $\theta(\omega)$ is the normalized resistance and $\chi(\omega)$ is the normalized reactance. Once the normalized specific acoustic impedance of the EMHR is known, the power into the EMHR is then computed by

$$W = \frac{|P|^2 \cos \vartheta}{2|\zeta(\omega)/A_D|}, \quad (5)$$

where P is the input pressure at the entrance of the EMHR, ϑ is the phase angle of ζ , and A_D is the cross-sectional area of the PWT. Note that the power into the EMHR is less than the

acoustic power of the incident plane wave since a portion of the incident acoustic power is reflected by the EMHR. The incident acoustic power is given by

$$W_{\text{in}} = \frac{W}{1 - |\mathfrak{R}(\omega)|^2}. \quad (6)$$

The experimental setup used to demonstrate operation of the integrated active liner system is shown in Fig. 10. Two PWTs, PWT₁ and PWT₂, are used to demonstrate the impedance tuning and energy harvesting performance of the EMHRs, respectively. The signal used in each of the PWTs is generated using the Bruel & Kjaer (B&K) pulse analyzer system and amplified with a Techtron 7540 power amplifier. A BMS 4590 compression driver in each tube uses the amplified signal to create the acoustic field. The EMHR backplate is comprised of an APC 850 piezoelectric disk on a brass shim. The dimensions of the PWTs are shown in Fig. 10 and the dimensions of the EMHR element used in each tube are given in Table II. The material properties for the piezoelectric backplates are provided in Table III. A total of four B&K 4138-A 1/8 in. microphones are used to simultaneously measure acoustic pressure in the PWTs. Two microphones, M_{IT} (impedance tuning) and M_{EH} (energy harvesting), monitor the acoustic pressure at the faces of the EMHR in PWT₁ and PWT₂, respectively. Two additional microphones, M_1 and M_2 , are used in PWT₁ to measure the normal specific acoustic impedance of the EMHR using the standard TMM.

The flyback converter in the energy harvesting block uses the standard topology presented in Fig. 7. Both capacitors, C_{rect} and C_{batt} , are 330 μF , and the inductance of the flyback transformer, L_{XRF} , is 6.8 mH. A complete discussion of the control electronics can be found in Ref. 18. The storage element is a 5 V battery comprised of four rechargeable AA Energizer cells in series. An ammeter placed in series with the battery monitors the magnitude and direction of

TABLE II. EMHR dimensions.

Dimensions of EMHR1	
Neck	Dimension (mm)
Radius r	2.42
Length t	3.16
Cavity	Dimension (mm)
Radius R	6.34
Depth L	16.4
Piezoelectric backplate	Dimension (mm)
Radius of the piezoelectric layer R_1	9.06
Thickness of the piezoelectric layer h_p	0.11
Radius of shim R_2	13.5
Thickness of shim h_s	0.22
Dimensions of EMHR2	
Neck	Dimension (mm)
Radius r	2.42
Length t	3.16
Cavity	Dimension (mm)
Radius R	6.34
Depth L	16.4
Piezoelectric backplate	Dimension (mm)
Radius of the piezoelectric layer R_1	11.75
Thickness of the piezoelectric layer h_p	0.14
Radius of shim R_2	17.25
Thickness of shim h_s	0.33

power flow of the storage element. The power for the transmitter is provided externally with a 9 V voltage source.

The switching array of the impedance tuning block is configured with four different electrical loads; open circuit, short circuit, 5 k Ω , and 500 Ω , where each electrical load provides a different acoustic impedance in accordance with

TABLE III. Materials properties of the piezoelectric backplate.

Piezoceramic APC850	
Young's modulus (GPa)	63
Poisson's ratio	0.31
Density (kg/m ³)	7700
Relative dielectric constant	1750
Piezoelectric strain constant d_{31} (pC/N)	-175
Shim (260 half hard brass)	
Young's modulus (GPa)	110
Poisson's ratio	0.38
Density (kg/m ³)	8530

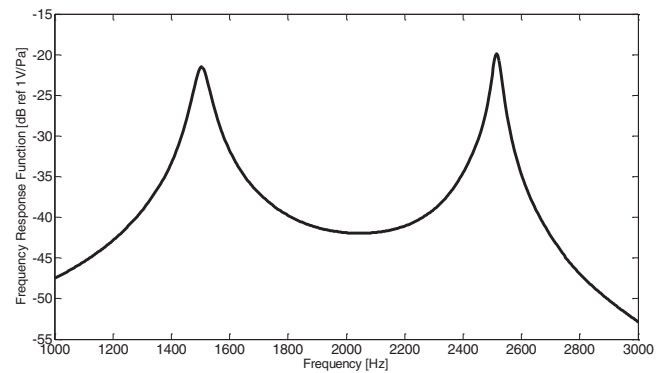


FIG. 11. Measured FRF (output voltage vs input acoustic pressure) of the EMHR in the energy harvesting block.

Eq. (1). These specific resistive loads were chosen to demonstrate the tuning range of the acoustically self-powered active liner. MAX 4544 single-pole, double-throw switches were used to connect the resistive loads to the EMHR when a logic value of 1 was sent by the ICS.

IV. EXPERIMENTAL VERIFICATION AND RESULTS

The first step in testing the active liner system was to determine the optimum acoustic signal to be used for energy harvesting. Turbomachinery noise in aircraft nacelles exhibit both tonal and broadband characteristics.¹⁹ For this demonstration, a single frequency excitation was chosen to emulate the tonal blade passage frequency component. For the EMHR in the energy harvesting block, the FRF between the output voltage and the input acoustic pressure, shown in Fig. 11, exhibited two peaks, which coincided with the two resonant frequencies of the EMHR.⁸ Due to the weak coupling between the piezoelectric backplate and the solid-walled Helmholtz resonator,⁸ the first peak is associated with the solid-walled Helmholtz resonator and occurs at 1.504 kHz. The second peak is associated with the resonant frequency of the piezoelectric backplate of the EMHR and occurs at 2.516 kHz. The excitation signal was then set to 2.516 kHz because it provided a larger electrical response for a given acoustic input. This maximizes the amount of energy harvested by the active liner.

As stated previously, for an EMHR element connected to a rectifier-capacitor circuit, a maximum amount of power will be harvested when the circuit is loaded with an optimal resistance value. The flyback converter is used to emulate a resistive load, and can be electronically tuned to the optimal resistance. For this experiment, the optimal resistance was found experimentally by replacing the flyback converter with a range of resistor values. The output voltage across each resistance was measured and the power was calculated using

$$P = \frac{V^2}{R}, \quad (7)$$

where V is the measured output voltage across the resistance R . The results of this test, shown in Fig. 12, indicate that the optimal resistance is approximately 20 k Ω . The flyback converter was then replaced, and tuned using Eq. (2). With a

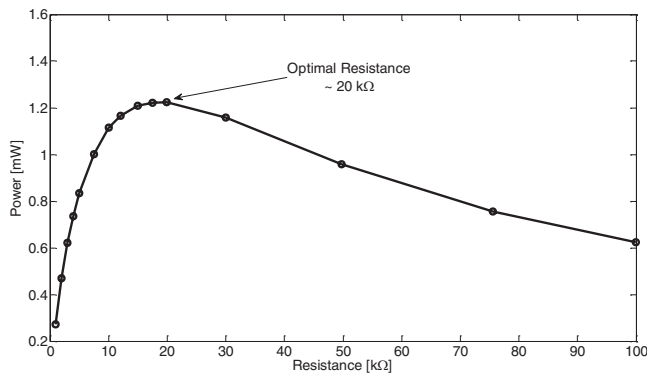


FIG. 12. Output power vs resistance for an EMHR element showing an optimal resistive load of 20 kΩ.

6.8 mH inductor and a switching frequency of 5 kHz, the duty cycle of the pulse width modulated (PWM) signal was set to 5.8% to achieve the optimal resistance of 20 kΩ.

Next, in order to determine if the self-powered active liner system was properly tuning the acoustic impedance of the duct, a control experiment was performed. Discrete loads, corresponding to the four loading conditions of the active liner (open circuit, short circuit, 500 Ω, and 5 kΩ),

were directly shunted to the EMHR in the impedance tuning block instead of the switching array. The pulse system was used to provide a broadband excitation in PWT₁ with a pseudorandom bandpass filtered signal from 300 Hz to 6.7 kHz. The acoustic signals needed for the TMM were recorded with M_1 and M_2 . A fast Fourier transform (FFT) was performed (1600 spectral lines, 3.5 kHz center frequency, 6.4 kHz span, and 300 ensemble averages) on the acoustic signals.

The normal specific acoustic impedance of the EMHR for the control experiment with the four loading conditions is shown in Fig. 13(a). The resistive loads effectively change the acoustic impedance of the piezoelectric backplate and thus shift the resonant frequencies of the EMHR. However, as shown in Fig. 13(a), when the EMHR was attached with the resistive loads, the tuning ranges of the resonant frequencies are restricted between the short- and open-circuit cases. As the resistance is increased, the resonant frequencies shift toward the open-circuit case as expected. By varying the electrical resistance attached to the EMHR, the control experiment shows that the second resonant frequency of the EMHR can be tuned to 212 Hz, from 2324 Hz (short-circuit

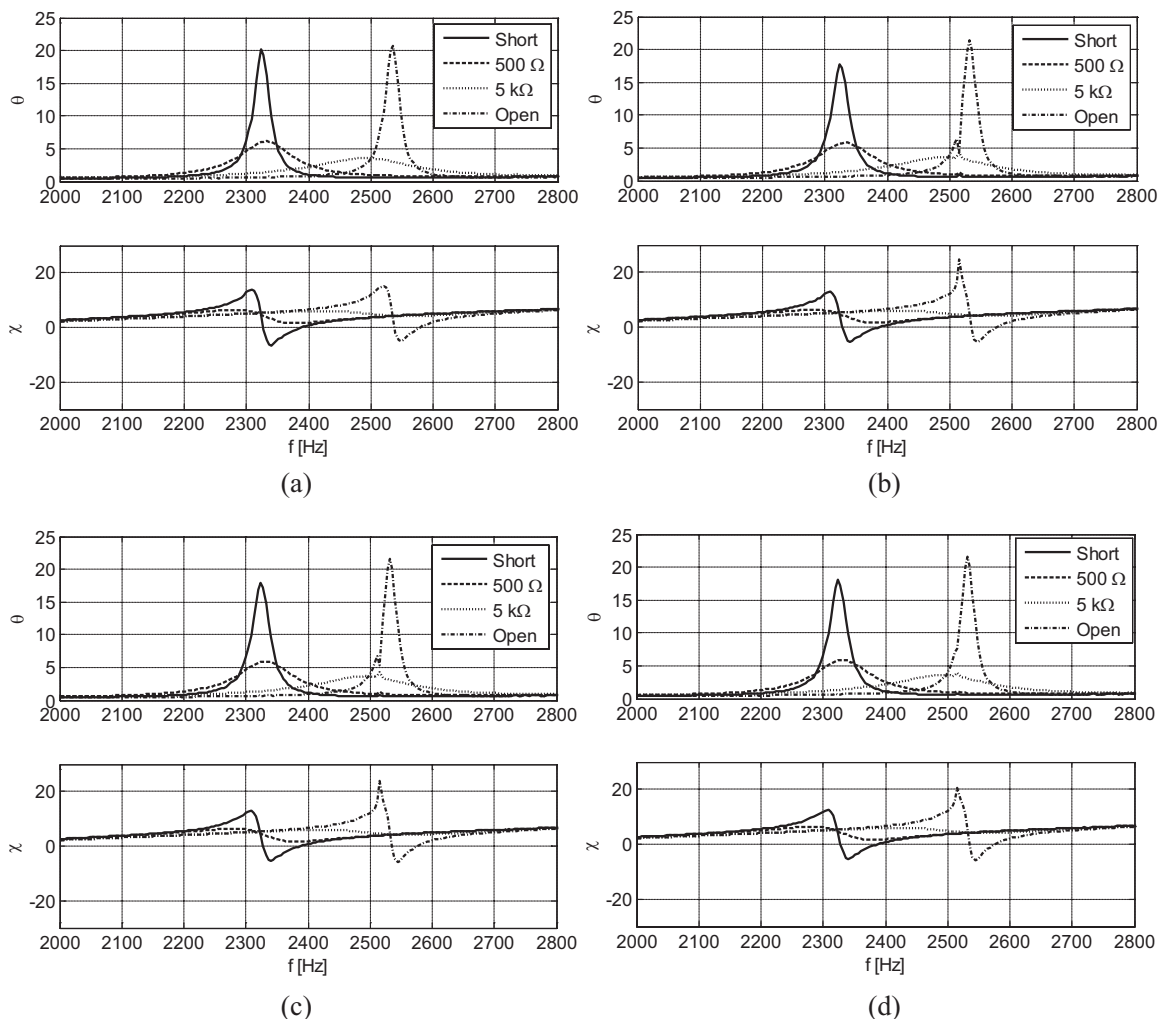


FIG. 13. Acoustic impedance of the EMHR for the (a) reference case, (b) positive battery current test case, (c) zero battery current test case, and (d) negative battery current test case.

TABLE IV. SPL and battery current from the active liner experiments.

	Case I	Case II	Case III
SPL (dB)	153.3	151.8	148.3
Current to the battery (μA)	680	10	-620

case) to 2536 Hz (open-circuit case). Moreover, resistive loads reduce the amplitude of the impedance peaks by increasing the system damping.

With the control experiment as a reference case, the acoustically powered switching array was used to verify the operation of the self-powered active liner system using wireless tuning. The complete liner system was tested under three different scenarios to demonstrate operation with different amounts of available energy. The three test cases were (I) harvested energy greater than required, and energy is stored, (II) harvested energy equal to required, and no energy is stored, and (III) harvested energy less than required, and energy must be provided by the storage element. Note that case III assumes that at some previous time energy was harvested and delivered to the storage element. For each of the three scenarios, the amount of available energy was regulated by adjusting the sound pressure level (SPL) of the single frequency tone inside PWT₂ and monitoring the ammeter in series with the battery. Positive current indicated that energy was flowing into the battery and that the harvested energy was greater than the requirements of the liner circuitry. Negative current indicated that the harvested energy was insufficient to power the liner, and stored energy must be used. Zero current occurred when the amounts of harvested energy and required energy were equal.

As with the reference case, the TMM was used to find the normal specific acoustic impedance of the EMHR for each of the test cases I–III, measured under the different resistive loading conditions (open circuit, short circuit, 500 Ω , and 5 k Ω). The same FFT settings were employed for consistency. The plots of the normal specific acoustic impedance of the EMHR for cases I–III are shown in Figs. 13(b)–13(d), respectively. The SPL needed to achieve each of these cases, as well as the battery current levels, is listed in Table IV.

Examination of the experimental results in Fig. 13 shows good agreement between the reference system and the three wirelessly tuned test cases. For all three test cases, the shift in the second resonant frequency of the EMHR was approximately 208 Hz from short- to open-circuit conditions, and is very close to the 212 Hz exhibited by the reference system. As shown in Fig. 13, the amplitude of the peak of the specific acoustic resistance of the EMHR with the short-circuit load was lower for these three test cases than for the reference. This discrepancy can be attributed to the finite resistance of the switches in the switching array of the impedance tuning block. Since the switch resistance is small (on the order of ohms), it was negligible for the other loads. On the other hand, the amplitude of the peak of the specific acoustic resistance of the open-circuited EMHR was system-

atically higher for the test cases, which employed the switching array circuitry. The exact reason for this phenomenon is not yet clear.

Examination of the specific acoustic reactance for all three of the test cases shows a small peak at 2.516 kHz, which is the frequency of the single tone used for energy harvesting in PWT₂. However, no peak is present in the reference case. One possible explanation for this peak is coupling of the acoustic signals between the PWTs. In order to power the liner electronics, the single tone signal used for energy harvesting required a relatively high SPL, on the order of 150 dB. It is possible that this high energy signal coupled with the broadband signals used for the TMM measurements in the neighboring tube and caused the peaks. This explanation seems reasonable since PWT₂ was off during the reference experiment, and the magnitude of the peaks decreased as the SPL in PWT₂ decreased for the test cases.

V. CONCLUSIONS

For the first time, an integrated, wireless, self-powered, active liner system has been demonstrated using EMHR elements for both acoustic impedance tuning and energy harvesting. Wireless operation of the integrated system was verified by demonstrating the ability to tune the liner using only acoustic inputs and wireless control signals.

The principles used to demonstrate the liner operation using only two EMHR elements can be expanded to encompass an array of elements. If more power is required or lower SPL is present, more than one EMHR can be used to power the liner. Each group of EMHR for power generation and tuning can be generalized as a module that is operated by a “master” controller. In this manner, the local liner impedance can be adjusted using an array of such modules. The independent operation of each module of the array improves the system robustness and reliability.

Finally, it should be emphasized that since this demonstration system was designed using mostly off-the-shelf components, future work should investigate the design of a custom receiver module and power converter to minimize both power demand and size.

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¹R. E. Motsinger and R. E. Kraft, “Design and performance of duct acoustic treatment,” in *Aeroacoustics of Flight Vehicles: Theory and Practice Volume 2*, edited by H. H. Hubbard (Acoustical Society of America, New York, 1995), pp. 165–206.

²Federal Aviation Administration, *Noise Levels for U.S. Certified and Foreign Vehicles* (Federal Aviation Administration, Washington, DC, 2001).

³D. Guicking and E. Lorenz, “An active sound absorber with porous plate,” *Trans. ASME, J. Vib., Acoust., Stress, Reliab. Des.* **106**, 389–392 (1984).

⁴S. Beyene and R. A. Burdisso, “A new hybrid passive/active noise absorption system,” *J. Acoust. Soc. Am.* **101**, 1512–1515 (1997).

⁵J. M. deBedout, M. A. Franchek, R. J. Bernhard, and L. Mongeau,

- "Adaptive-passive noise control with self-tuning Helmholtz resonators," *J. Sound Vib.* **202**, 109–123 (1997).
- ⁶X. D. Jing and X. F. Sun, "Experimental investigations of perforated liners with bias flow," *J. Acoust. Soc. Am.* **106**, 2436–2441 (1999).
- ⁷S. Horowitz, T. Nishida, L. Cattafesta, and M. Sheplak, "Characterization of compliant-backplate Helmholtz resonators for an electromechanical acoustic liner," *Int. J. Aeroacoust.* **1**, 183–205 (2002).
- ⁸F. Liu, S. Horowitz, T. Nishida, L. Cattafesta, and M. Sheplak, "A multiple degree of freedom electromechanical Helmholtz resonator," *J. Acoust. Soc. Am.* **122**, 291–301 (2007).
- ⁹N. E. duToit, B. L. Wardle, and S. G. Kim, "Design considerations for MEMS-scale piezoelectric mechanical vibration energy harvesters," *Integr. Ferroelectr.* **71**, 121–160 (2005).
- ¹⁰M. Ferrari, V. Ferrari, D. Marioli, and A. Taroni, "Modeling, fabrication and performance measurements of a piezoelectric energy converter for power harvesting in autonomous microsystems," *IEEE Trans. Instrum. Meas.* **55**, 2096–2101 (2006).
- ¹¹J. Jun, B. Chou, J. Lin, A. Phipps, X. Shengwen, K. Ngo, D. Johnson, A. Kasyap, T. Nishida, H. T. Wang, B. S. Kang, F. Ren, L. C. Tien, P. W. Sadik, D. P. Norton, L. F. Voss, and S. J. Pearton, "A hydrogen leakage detection system using self-powered wireless hydrogen sensor nodes," *Solid-State Electron.* **51**, 1018–1022 (2007).
- ¹²F. Liu, A. Phipps, S. Horowitz, T. Nishida, L. Cattafesta, and M. Sheplak, "Acoustic energy harvesting using an electromechanical Helmholtz resonator," *J. Acoust. Soc. Am.* **123**, 1983–1990 (2008).
- ¹³S. Kadirvel, F. Liu, S. Horowitz, T. Nishida, K. Ngo, L. Cattafesta, and M. Sheplak, "A self-powered wireless active acoustic liner," presented at the 12th AIAA/CEAS Aeroacoustics Conference, Cambridge, MA, 2006.
- ¹⁴R. A. Mangiarotti, "Acoustic-lining concepts and materials for engine ducts," *J. Acoust. Soc. Am.* **48**, 783–794 (1970).
- ¹⁵J. Chen and K. D. T. Ngo, "Alternate forms of the PWM switch model in discontinuous conduction mode," *IEEE Trans. Aerosp. Electron. Syst.* **37**, 754–758 (2001).
- ¹⁶"Impedance and absorption of acoustical materials using a tube, two microphones, and a digital frequency analysis system," ASTM-E1050-98, ASTM International, 1998.
- ¹⁷T. Schultz, M. Sheplak, and L. N. Cattafesta, "Uncertainty analysis of the two-microphone method," *J. Sound Vib.* **304**, 91–109 (2007).
- ¹⁸R. J. Taylor, "Optimization of a discontinuous conduction mode flyback for acoustical energy harvesting," MS thesis, University of Florida, Gainesville, FL (2004).
- ¹⁹M. J. T. Smith, *Aircraft Noise* (Cambridge University Press, Cambridge, 1989).